

Aaron Danen

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Education

University of California, Davis – B.S. in Computer Science and Engineering, 3.97 GPA Expected June 2027

Coursework: Advanced Compilers (CS6120), Computer Architecture, Operating Systems, Programming Languages

Honors: Dean's List, Tau Beta Pi Honor Society

Skills

Languages: C, C++, Python, LLVM IR, x86 Assembly

Tools: git, clang/gcc, cppcheck, CMake, Make, Linux, Cygwin

Libraries: LLVM, OpenGL, Pandas, Matplotlib, Numpy

Experience

Compilers Intern, Hewlett Packard Enterprise June 2026 - September 2026

- Modified preprocessor, parser, and dataflow analysis of open source static analysis tool Cppcheck to support Nonstop C/C++ extensions such as optional function arguments, multiple size pointers, and compiler intrinsics
- Discovered and reported real memory safety bugs in Nonstop software using modified Cppcheck analyzer
- Iterated with QA engineers to develop CLI features and optimal workflows for analyzing Nonstop software

Data Engineering Analyst, UC Santa Cruz, Dept. of Astrophysics Aug 2022 - Sept 2023, Oct 2024 - Aug 2025

- Automated data retrieval and preprocessing pipelines for VERITAS and NICER telescope data using Python and Pandas, reducing manual labor by 10+ hours per week for researchers seeking up-to-date measurements
- Automated analysis and aggregation of data from 4 observatories, contributing to a \$90,000 grant proposal
- Surveyed and met users visualization needs by enriching the data through statistical methods and using Matplotlib
- Leveraged existing NASA CLI tools into a Python interface with shell scripts and string processing algorithms

Open Source Contributions

LLVM, Compiler Framework github.com/llvm/llvm-project/

- Fixed a crash in Windows and WebAssembly exception handling code generation by detecting malformed "catchpad" instruction arguments and rewriting the IR to a valid form that traps if an exception is thrown
- Strengthened clang-repl UI/UX by implementing a %help command and clarifying error messages and documentation

Cppcheck, Static Analyzer github.com/cppcheck-opensource/cppcheck/

- Reduced analyzer noise by extending escape analysis to handle function pointers, lambdas, and callable objects
- Eliminated false positive uninitialized memory reads by modeling the semantics of compound assignment operators
- Contributed stubs for GCC bit operation builtin function

BJET_MCMC, Statistical Analysis Software github.com/ohervet/Bjet_MCMC

- Improved runtime by 20% by running tests on a OpenHPC cluster to optimize Monte Carlo Markov Chain parameters
- Authored performance and optimization documentation page to save users computation time
- Contributed a feature that fits an exponential decay function to the Markov Chain convergence and reports to users

Projects

Compiler Optimization Library github.com/aadanen/baleen

- Optimized BRIL programs to execute less instructions by implementing classical compiler optimizations in a library
- Developed an extensible and useful product by leveraging object oriented patterns, a flattened AST, and CLI utilities

CHIP-8 Interpreter github.com/aadanen/chip8

- Designed a retro emulator that supports all standard ROMs by mimicking various sets of historical quirks

Activities

UC Davis Competitive Programming Club President

California All-State High School Honor Jazz Band